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RCT evidence on differential impact of US job training programmes by pre-training employment status

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ABSTRACT

While the recent evidence around the effectiveness of job training programmes has been generally positive, there has been little evidence regarding the differential impact in the U.S. by pre-training employment status. The city of New Orleans implemented a job training programme funded by the U.S. Department of labour Workforce Innovation Fund. They screened potential trainees for suitability and likelihood of completing and benefiting from training. They selected and developed curriculum with input from local firms. The city implemented the training as a randomized controlled trial. In this paper, we evaluate the impact of the training on employment, earnings, and industry of employment, both overall and separately by baseline employment status. We find overall positive effects for earnings, driven by those who did not have a job at the time of training enrolment. We also find positive employment effects for short-term unemployed workers. Individuals in the treatment group were more likely to work in the target industries, suggesting that the gains were happening through the gains in industry-specific human capital.

1. Introduction

Labour markets have become increasingly focused on knowledge-based industries, which require higher levels of technical skills. There is a resulting widening of the gap between the skills needed in the marketplace and those available from existing workers, with companies unable to fill vacancies and grow at the rate which they would like. As a result, the wage disparity between high- and low-skilled workers has grown (Acemoglu, 2002). Many workers with few or obsolete skills face diminishing demand, and worsening labour outcomes. These forces have combined to increase the polarization of job quality and income, leading to widening inequality (Autor and Dorn, 2013; Autor et al., 2006; Goos and Manning, 2007; Goos et al., 2014).

Providing effective pathways for unemployed and underemployed workers into sustainable and sustaining employment is a critical policy issue to combat both poverty and inequality. Active Labour Market Programmes (ALMP) provide potential interventions to improve worker outcomes, and training programmes are a common type of ALMP (Brown and Koettl, 2015; Kluve, 2010). These educational programmes have the potential to increase the human capital of these workers, better positioning them in the workforce landscape with in-demand skills. There are several types of programmes aimed at workforce development through increasing human capital, including formal post-secondary education (four-year and two-year college), vocational and career education, apprenticeships, job training programmes, and on-the-job training. These differ in the funder, the provider, and the format of the training. These types of programmes have shown promising returns in some settings, albeit not in every instance. Publicly-subsidized training programmes are typically medium-length (often between one week and one year) and have training provided by various entities, including community colleges, private training organizations, employers, and the government itself. Such programmes have existed in the United States and Europe for several decades and have a robust research base. The existing literature has found that the programmes can be effective, but are not always so (Van Horn et al., 2015; Fortson et al., 2017; Brown and Koettl, 2015).

Two of the issues that hurt the effectiveness of training programmes are high attrition and mismatches between skills provided and local labour demand. To that end, the New Orleans Office of Workforce Development (OWD) designed and implemented *Career Pathways*, funded by a US Department of labour (DOL) Workforce Innovation Fund (WIF) grant. WIF grants are authorized under the Workforce Innovation and Opportunity Act (WIOA) passed in 2014. Career Pathways offered training for unemployed and underemployed individuals relevant for medium-

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skilled jobs in the advanced manufacturing, health care, and information technology (IT) sectors. OWD purposefully chose the industries and developed the curriculum in coordination with local firms, critical features of successful programmes referred to as demand-driven training. Further, to combat programme attrition, potential trainees were screened by various profiling tools before they were allowed entry into the randomization pool and thus into training. For evaluation purposes, the programme was implemented as a randomized controlled trial (RCT).

In this paper, we examine new and important aspects of the impact of the Career Pathways programme. In Baird et al. (2019), we evaluated the Career Pathways and found it to lead to increased earnings, especially for those unemployed at entry into the programme. We add to those findings and to the overall literature on effectiveness of job training programmes in several ways. First, while we are examining the same intervention as in Baird et al. (2019), we obtained more data than used in that report, expanding from that analysis of 13 training cohorts to 20 cohorts in this paper, as well as examining post-training outcomes for an additional six months. Additionally, the programme allowed for persons randomized into a control group to enter later randomization pools, resulting in some individuals moving from the control group to the treatment group in later quarters. We account for these individuals in a different manner than in Baird et al. (2019) (particularly important given our longer timeline and 7 additional later cohorts), eliminating the risk of bias from the selection of motivated repeaters out of the control group.

This paper also differs from Baird et al. (2019) by focusing on an important moderator of the outcomes, the individual's baseline employment status (has a job, short-term unemployed, long-term unemployed). Baird et al. (2019) grouped short and long term unemployed together in evaluating this moderator, and did not examine it across all the outcomes examined here. Understanding treatment effect heterogeneity is important, as it helps to shed light on how trainees are or are not benefiting, which can be used for targeting of interventions, developing appropriate post-training supports (in areas where they are not succeeding), and building best practices.

We find that the training is effective in terms of increasing earnings: the intent-to-treat (ITT) estimate of $436/quarter yields about a 12 % increase over the control group; the local average treatment effect (LATE) estimate using an instrumental variable estimator for those who attended training is $1054/quarter for about a 30 % increase over the control group. The effectiveness of the programme is largest for short-term unemployed trainees, a group of particular interest. For them, the ITT estimate is $1450. This group also has positive and significant estimates for the impact of the programme on the probability of being employed. We also find that trainees were more likely to work in target industries, although this effect does not seem as differentiated amongst employment groups.

The paper proceeds as follows. Section 2 describes the prior literature regarding job training programmes for adults. Section 3 discusses the context, provides a conceptual framework, and presents the data sources used in this analysis. Section 4 describes the methodology used in the paper, while Section 5 presents the results of our analyses. Section 6 concludes.

2. Prior literature

The literature on evaluations of ALMPs and job training programmes specifically have a long tradition in the economics literature. Job training programmes have generally been found to yield positive outcomes, with gains to employment and earnings for participants (Card et al., 2010; McCall et al., 2016; Council of Economic Advisers, 2019; Brown and Koettl, 2015; Roder et al., 2008).

This study focuses on evaluating a programme in the United States. The U.S. Department of labour provides funding for innovative training programmes through the Workforce Innovation Fund (WIF), authorized by the Workforce Innovation and Opportunity Act (WIOA) signed into law in 2014 and its predecessor, the Workforce Investment Act (WIA). WIF programmes are of two types: Adult programmes and Dislocated Workers programmes. Individuals may qualify for the Dislocated Workers programme in several ways, including but not limited to being laid off (either by facility closing or otherwise) while being unlikely to return to work in that industry, being self-employed but without significant work opportunities, and or having relied on the income of another family member who no longer provides such income (Department of Labor, 2017). Adult programmes make no such requirement. Our study evaluates an intervention in the Adult programme and not the Dislocated Workers programme. We focus on differential impacts by pretreatment employment level. However, we are unable to identify a subpopulation that meet the same classification as those in the Dislocated programme. Nevertheless, while we focus our review on Adult programs and similar type programs in Europe, we briefly discuss some literature on Dislocated Worker training programmes as interesting comparisons.

The most comprehensive study of training under the WIA–funded programmes is reported in Fortson et al. (2017), a follow-up on McConnell et al. (2016). Fortson et al., report on an RCT design similar to our study but did so across 28 workforce investment boards' training programmes. They found statistically significant average increases of over $600 a quarter in earnings from treatment assignment after training. However, these results only are evident in the survey responses of earnings; using administrative data, there is a smaller, positive, and not statistically significant effect on earnings from training assignment. They also found that providing training was not cost effective, which they explain as partially due to a substantial fraction of members of the control group finding alternative ways to fund participation in the same training programmes (participation rate of about 40 % in job training, compared to around 50 % of the treatment group. In our study, while control persons could seek training on their own (including other programmes from the same training providers), they were not eligible for participating in the WIF training programme cohorts even if they paid for it themselves. However, Fortson et al., found that providing intensive services without training was cost effective. They also found no positive returns for individuals in the Dislocated Worker programmes.

Earlier evaluations of WIA also provide a useful benchmark for the present program, although they often relied on quasi-experimental methodology. Heinrich et al. (2013) evaluated WIA training programmes using quasi-experimental design methods in twelve states for up to four years after the start of the programmes. They found moderately sized, positive, and statistically significant treatment effects on employment and earnings. For example, they estimated quarterly earnings increases of $591 for women and $419 for men (around a 25 % and 15 % increase over base earnings for women and men, respectively) for the WIA Adult programmes. Employment rates had an average of around six percentage-point increases. However, they found no returns in the WIA Dislocated Worker programs, available to those that had been laid off. Thus, those potentially most in need of the programme benefitted the least. These results are similar to Andersson et al. (2013), who in investigating a different set of WIA programmes found similar effects: for the adult classification, earnings effects were around a $300-$600 increase in quarterly earnings overall in the first three years after training, with the estimated increase for the third year being $1300-$1700, while the increase in employment rates was around two percentage points. However, as in Heinrich et al. (2013), they found a worse outcome for trainees in the Dislocated Workers program, finding significant average earnings losses over the first three years (although with gains in the third year). Hollenbeck (2012) found a postsecondary CTE training programme yielding around a $1500 increase in quarterly earnings. Andersson et al. (2013) noted that it remains a puzzle why the Dislocated Worker programme tends to be less successful in generating increased income and employment, compared to the Adult program, especially considering the services provided are very similar and the observable characteristics of the trainees do not differ nearly as much as their impact estimates.

One characteristic of job training programmes that has been shown to be effective is having it be demand-driven: training programmes that are highly integrated with the local labour market and private sector. In fact, it is one of the changes made in the transition from WIA to WIOA training programmes (Fortson et al., 2017, p.5). Demand-driven strategies consist of implementing services and activities—including job training—that focus on the needs of specific sectors. In other words, these training programmes are based on market-aware strategies that identify sectors that have unmet needs for workers and provide training to low-income workers to acquire the needed skills to fill the available positions (Roder et al., 2008). The results obtained in demand-driven training programmes are encouraging. For example, Maguire et al. (2010) evaluated the employment and earning impacts of three demand-driven training programmes and found gains not only from working more hours but also from higher wages. Van Horn et al. (2015) provide a more in-depth review of this literature.

Although this study focuses on a U.S.-based programme, job training and ALMPs are of course not unique to the U.S. While it is beyond the scope of this paper to outline all of the similarities and differences between job training programmes in the U.S. and in Europe, the basics of publicly-subsidized job training programmes in the two areas are similar enough to warrant discussion of European job training programmes (Méndez and Sepúlveda, 2016; McCall et al., 2016; Misko, 2006). In a meta-analysis of European AMLPs, Kluve (2010) found overall modest positive impacts across 70 job training programme evaluations (see also Kluve and Schmidt, 2002). Evaluations of specific job training programmes in Europe have at times been very favourable in terms of increased wages and employment (Winter-Ebmer, 2001; Leetmaa and Vörk, 2004; Biewen et al., 2014; Dauth and Toomet, 2016), while others have been negative (Bolvig et al., 2003; Hujer et al., 2006). Training effectiveness depends on worker and program characteristics and alignment (Zwick, 2015; Biewen et al., 2014). Cockx (2003) looked specifically at a programme aimed at unemployed workers in Belgium and found, while transitions into unemployment increased during training, it decreased even more after the end of training, providing evidence in favour of job training programmes for unemployed workers. (see also Brodaty et al., 2002; Biewen et al., 2014 for similar findings).

This paper contributes to the research in two important ways. First, it adds to research on the effectiveness of job training programmes using the rigorous methodology of an RCT. Most of the literature on the returns to training programmes in the last few decades, as described above, have relied on quasi-experimental methods, with the primary exception being the Fortson et al. (2017) evaluation. Nonexperimental methods to evaluate social programmes can be problematic and fragile, as discussed in LaLonde (1986). RCTs do have their limitations: they are at times "black boxes," showing whether an intervention was successful without necessarily shedding light on why or how it succeeded or failed, thereby limiting replication (see List and Rasul, 2011, Heckman and Smith, 1995, and Heckman, 2000 for a discussion of these limitations). However, this is no more true for RCTs than it is for quasi-experimental methods, and experimental estimates continue to be the gold standard in estimating causal effects.

Our second contribution is to take a closer look at heterogeneity in treatment effects depending on labour history of the trainee. We separate the sample into three groups: those with jobs at the start of training, those who did not have a job at the start of training but had held one within the prior six months, and those who had not held a job within the prior half year. In addition to examining employment and earnings, we look at placement into the target industries. The placement into the target industry sheds light onto the question of whether the participants are developing skills that are currently demanded in the local labour market. This is typically not studied; however, Fortson et al. (2017) looked at this in 28 WIA programs, and found the treatment group were not more likely to work in the target occupations. In the European context, Biewen et al. (2014) showed that short-term job training programmes in Germany were most effective for those who we would classify as short-term unemployed.

3. Context

3.1. Conceptual framework

We are interested in understanding the impact adult job training has on employment and earnings, and how that effect may differ depending on employment history. There may be several pathways through which participation in the training programme may affect these labour outcomes. The primary hypothesized pathway is that the training will increase worker human capital and thus productivity, which in turn should increase employment probabilities and expected earnings (Becker, 2009). There is evidence that training can increase not only "hard skills"—technical knowledge and expertise—but also can build "soft skills" (Acevedo et al., 2020), which are highly valued by employers (Deming, 2017). Additional potential favourable mechanisms for affecting employment and earnings include signalling their ability and grit through training completion and credentialing (Hungerford and Solon, 1987; Bushway and Apel, 2012); networking with instructors, other students, and the schools (Forret, 2018; Flórez et al., 2018); and improved match quality between the employee and employer (Zhang et al., 2021; Brown and Koettl, 2015). A potential unfavourable mechanism is temporary (full or partial) removal from the labour market during training (Ham and LaLonde, 1996). On net, we hypothesize that in a well-targeted, designed, and executed job training program, the positive mechanisms will dominate.

We explore the possibility that baseline employment status is an important moderator of the success of the training. We consider three groups of trainees based on pre-training employment status: those who had a job at the start of training (*job at start*); those who did not have a job at the start of training but had held a job within the prior half year (*short-term unemployed*); and those who had not held a job in over half a year (*long-term unemployed*).

The first potential pathway for this moderator is differences in the opportunity cost of training for a trainee. Hujer et al. (2006) found that participation in vocational training in Germany may prolong unemployment duration. Those that already have a job or receive a good job offer during the course of the training may be more likely to drop out of the training. From this, we hypothesize that the *job at start* group would be the least likely to complete training or focus completely on the training, followed by the *short-term unemployed*. In the control group, those with favourable employment history will be more likely to have a stronger positive trajectory in their careers as well. Together, these imply smaller treatment effects for the *job at start* group, because treated persons in this group will be more likely to drop out and control individuals will be more likely to have better labour outcomes. The same may also be true, albeit to a lesser extent, for the *short-term unemployed* individuals.

A second potential pathway for this moderator suggests that the relationship between employment status and training impact might go in the opposite direction. Employment history will be positively correlated with unobserved, time-invariant levels of ability and persistence. For this reason, the long-term unemployed may be the least likely to complete training, thereby decreasing treatment effects. Further, their experience with training and their ability to leverage increased human capital to new job offers may differ by these same abilities, and firms may prefer those with recent employment history, together leading to higher treatment effects for those with a job at start or the short-term unemployed. This implies that labour history will be related to future employment transition probabilities (Dorsett and Lucchino, 2018; Paul, 2015). In line with these counteracting effects, research has shown that training effectiveness has been shown to vary depending on pre-training employment status and unemployment duration (Biewen et al., 2014). Even long-term unemployed workers can benefit from such programmes (Brodaty et al., 2002).

3.2. Description of the Career Pathways program

The Career Pathways training programme in New Orleans, Louisiana recruited over 600 individuals and offered training to over 300 individuals across 25 cohorts in one of three training pathways (advanced manufacturing, health care, and information technology) occurring between 2016 and 2018. OWD recruited potential candidates using several methods, including local One-Stop Centers where individuals came for help with employment and public programmes such as Unemployment Insurance and SNAP, fixed tablet stations placed in targeted local hotspots throughout the city, targeted outreach through paid advertising and online marketing campaigns (e.g., Facebook, Craigslist, Twitter, and radio ads), and existing community partnerships.

Interested candidates for training were then screened for inclusion in the study. The screening process differed slightly across cohorts, but overall addressed the goal of identifying candidates more likely to persist in the training program, through literacy and numeracy readiness, as well as dependability and availability. The screening process typically included four components: (1) attendance at a mandatory orientation, (2) drug testing (and at times a criminal background check), (3) completion of a relevant assignment or test to determine basic literacy and numeracy (such as the Test of Adult Basic Education, or TABE) and (4) completion of a structured and scored 45 min interview with OWD about their work readiness.

OWD used these four tools to gauge candidates' interest in the career pathways and likelihood for successfully completing the training programmes. Those deemed likely to succeed were placed into the randomization pool, while those unlikely to succeed were directed to other government programmes and benefits. While selection of trainees is always a consideration of any training program, this rigorous level of determination and selection is uncommon. In Fortson et al. (2017) review of 28 programs, they found that sites used assessment tools, including the TABE, but did so for information on how to support the trainee and not as selection criteria. Fortson et al., recommend that these kinds of assessment tools be used to select trainees in much the way that Career Pathways did, which may be related to the better findings in this study.

Candidates approved for training after the screening were invited to attend an additional meeting, wherein they were informed that in order to qualify for training, they had to sign a consent form in which they agreed to be part of the study whether or not they were randomly assigned to the treatment status. For those who consented, half were randomly selected to be invited to the training program, and half were not. Randomization was done using random numbers within strata defined by gender, annual income in the prior year being above or below $5000, employment status at that time, and age being 35 years or older. Training typically started within a week after randomization so as to minimize attrition at this stage.

The training was typically structured as in-classroom instruction, for 20 h across five days a week for a total of two months. In almost every cohort, individuals were then offered an additional two months of training to build on the first round. The training programme represented a coordinated effort between OWD and the training providers, which included a local community college, established professional training providers, and local firms, including a health network and Goodwill. OWD engaged with potential employers through Trade Advisory Committees, meeting quarterly to obtain an understanding of their workforce requirements and hiring needs and to develop the training. At the end of each 2-month round of training, candidates in most cohorts were invited to test for at least one industry-based credential in the form of a certification. The medium-length duration of the training served both as a challenge (skills desired may take significantly longer to develop) and an opportunity (workers were more likely to be able to afford the opportunity cost of working less during a shorter training), and was chosen to strike a balance between these two competing issues.

We are unable to observe whether individuals in the control group opted into similar training programs outside of our study during the timeframe. We are thus assuming that the control represents business as usual, which will allow for the possibility that some members of the control group received training that they obtained elsewhere. Assuming an average positive effect of training in alternative programs, our ITT and LATE estimates of the program impact compared to business as usual will tend to be smaller than estimates of the impact of training relative to *not* receiving any training in any other settings.

3.3. Data

Our data comes from two sources. First, in order to enter a randomization pool, individuals had to fill out a brief baseline survey with questions about their demographics and current employment and earnings. They also provided their social security numbers (SSN) and consented to share their employment outcomes from the state of Louisiana. This enabled the second data source, which contains employment, earnings, and industry records merged on SSN from the Louisiana Workforce Commission (LWC) for each individual for each employer that was reported to the state for each quarter between and including the first quarter of 2014 and the first quarter of 2019, for 21 total consecutive quarters of data for each individual. 68.2 % of these quarter-person observations were before the start of training, 8.5 % of quarter-person observations were during the training period of the cohort they were randomized within, and 23.4 % of quarter-person observations were after the end of the training period for the cohort they were randomized within. While the exact time frame differs depending on when they were randomized, on average we observe 16 quarters before the end of training (minimum, 14, maximum 20) and 4 quarters after the end of training (minimum, 1, maximum 7).

The two data sources provide somewhat difference perspectives on employment and earnings history. The baseline survey question about current work status at time of randomization does not exclude self-employment or employment in jobs not covered by the Louisiana unemployment insurance system, such as farm labour, work in the informal labour market, or "under-the-table" employment not reported to the state, and is with respect to the moment, and not employment over the prior quarter. Similarly, the survey question about annual income does not exclude earnings from these same jobs. On the other hand, the LWC data reflects whether the individual was employed in a covered job at some point during a particular quarter but does not identify employment at the time of randomization. We use both sources of information to measure the prior labour history of applicants, use the survey for additional control variables, and use the LWC data to measure the outcomes.

We define a cohort as a group of individuals entering the same randomization pool for a specific training programme at the same time. There were 25 training cohorts across the Career Pathways programmes. The first was a pilot cohort and not randomized, and so is not included in the analysis. We also drop the next two cohorts, given large changes in aspects of Career Pathways (in particular, in how individuals are recruited and screened) between these first two cohorts and the subsequent 22 cohorts. The last two cohorts are not included in this study because we were unable to acquire data on their post-training outcomes. This leaves us with 20 cohorts on which to evaluate the effectiveness of the programmes. OWD allowed individuals that were randomized into the control arm to re-enter later randomization cohorts. Some control individuals did so, and some of these were later randomized into treatment. For the purposes of this paper, we define treatment status according to their first treatment assignment, even if a control individual was later randomized into treatment for a subsequent cohort. As we discuss in Section 3, this controls for non-random selection biases that could arise from reclassifying treatment status following later treatment assignments.

Table 1 provides information on the number of cohorts, individuals, and observations for different pathways in the analytic sample. Overall, 197 people were first assigned to be in the treatment group, and 192 in

Table 1
Training Pathways counts of people and person-quarters.

Pathway	Area	Cohorts	Number of individuals assigned treatment	Number of individuals assigned control	Post-training quarterly observations
Advanced manufacturing	Electrical	5	77	67	876
	Pipefitting	1	5	6	22
	Welding	1	11	8	57
Health care	Medical billing and coding	4	32	32	176
	Patient access representative	1	19	20	195
Information technology	Information technology	8	53	59	575
Total		20	197	192	1901

Table 2
Treatment compliance.

		Attended Training 1?		Completed Training 1?		Attended Training 2?		Completed Training 2?		
		No	Yes	No	Yes	No	Yes	No	Yes	Total
Assigned to Treatment?	No	166	26	170	22	185	7	187	5	192
	Yes	35	162	71	126	153	44	162	35	197
	Total	201	188	241	148	338	51	349	35	389

Note: Attended training refers to whether the individual attended at least one class in the 2 month training. Completed training refers to whether the individual completed the given round of training.

the control group. This excludes individuals in later cohorts that were in an earlier cohort's control group, as discussed. However, given treatment assignment in every cohort is random, these excluded observations from later cohorts will be randomly taken out of both the treatment and control, ending with balance both in terms of numbers of people in each group and demographics of each. About half of the individuals were in the advanced manufacturing pathway, and a quarter in each of health care and information technology. Overall, there are almost 2000 post-training quarterly observations in our analytic sample.

We also have measures of whether individuals attended at least one training class in the first and second round of training and whether they completed that first and second round of training (Table 2). 18 % of individuals assigned to training (35 people) did not show up to even one class. There is also non-compliance with initial treatment assignment amongst the control group. This happens if an individual in the control group has all three of the following occur: they (1) return to the randomization pool for a subsequent cohort, (2) are randomized into treatment, and (3) attend at least one training session. Here, the level of non-compliance also is around 18 %. These non-compliances in both direction will lead to under-estimates of the causal effect of training on employment outcomes. We not only estimate ITT models based on initial assignment, but estimate a LATE model using whether the individual attended at least one training class as the treated group. 79 % of those that attended at least one class completed the first round of training. Additionally, while a majority of the treatment group attended and completed the first round of training, only a minority opted into participating in the second round of training, with fewer still completing that training.

A further complication is that not all SSNs in the baseline surveys were matched onto LWC records. There could be three reasons for our missing their employment data from this complication: (1) they might in truth never have worked in the state of Louisiana between January 1st, 2014 and March 31st, 2019; (2) they might only have worked in unreported jobs; or (3) there may have been an error in the SSN from the baseline survey (either in what was provided or in the digitization of it). We strongly suspect that most if not all cases are due to the last scenario of incorrect SSNs.[1] This represents 28 individuals that are not included in the analysis in addition to the 389 we include. Although there is near-zero difference in probability of treatment status by missingness and no statistical relationship, we generate and use non-response weights for the remaining subjects to account for these individuals.

Table 3 presents the baseline characteristics of the sample we use for this analysis, the employment category examined as a key moderator of the impact, and the employment and earnings from the LWC data prior to training and after training. We classify individuals into one of three employment groups. *Job at start* are individuals who on the baseline survey reported currently holding a job. These account for just over half of the individuals. *Short-term unemployed* are individuals who reported not having a job in the baseline survey, but for whom we observe reported earnings in the LWC data at some time in the same or prior quarter (chosen as a window to reflect not having been out of employment for a significant period, which also aligns with the unemployment benefits eligibility window). These account for just over one quarter of individuals. *Long-term unemployed* are individuals who reported not holding a job in the baseline survey and for whom we do not observe earnings in any job within the past two quarters. These account for just over one fifth of the sample.

We find that there is no statistical difference between the treatment and control group on any of the baseline dimensions examined in Table 3 (the p-value of a *t*-test of the difference in means is never below 0.1), and that the effect sizes (i.e., difference divided by pooled standard deviation) are quite small. Thus, while we will covariate-adjust for all of these variables in the impact regression to increase precision, they are not necessary to correct for imbalances between treatment and control groups. Further, there appears to be proper randomization within the employment groups. The table also shows the attributes of the sample: that they are predominantly Black individuals, of various ages, and have diverse labour backgrounds, with almost half not holding a job at the

[1] The LWC data covers a period of more than five years during which we found no matches for these unmatched individuals, and the set of individuals with no match in LWC did not change across waves of later LWC data acquisitions. If these individuals had correct SSNs and were merely missing because they had not worked at all to date, we might have expected subsequent data draws to include some of these SSNs as they finally move into paid employment. But this is not the case, providing suggestive evidence in support of the SSNs being incorrect. Also, 33% of respondents who were not in the LWC data reported in their baseline survey at time of randomization that they had partial or full employment in the month prior to the randomization, and 51% of the same group mentioned that they had personal income over $5,000 in the last year.

Table 3
Summary statistics and baseline equivalence.

	Control (N=948)	Treatment (N=953)	Difference	Std. Error of Diff.	Effect Size
Baseline characteristics					
Male	0.522	0.570	-0.048	0.164	0.096
Over 35 years old	0.535	0.584	-0.050	0.046	0.100
Prior year's income>$5000	0.620	0.638	-0.018	0.064	0.037
Black	0.886	0.921	-0.035	0.033	0.119
Age	39.066	40.160	-1.095	1.212	0.097
Proportion of prior quarters employed	0.614	0.625	-0.012	0.045	0.030
Average prior quarterly earnings	3815	3480	335	491	0.084
Employment group					
Job at start	0.517	0.504	0.013	0.051	0.026
Short-term unemployed	0.264	0.300	-0.036	0.044	0.081
Long-term unemployed	0.219	0.196	0.023	0.051	0.057
Outcomes					
Employed (post training periods)	0.639	0.690	-0.051	0.048	0.108
Quarterly earnings (post training periods)	3567	3909	-343	470	0.083

Note: Proportion of quarters employed and average quarterly earnings are based on LWC records for 0.5 years to 2.5 years prior to the start of training. Standard errors are clustered at the cohort by treatment status level. None of the associated p-values are significant at the 10 % level. Effect sizes are the difference divided by the pooled standard deviation.

Table 4
Baseline characteristics between employment groups.

	Job at start (N=200)	Short-term unemployed (N=110)	Long-term unemployed (N=79)	p-value for diff. between groups
Treat	50.5%	52.7%	48.1%	0.821
Female	47.5%	51.8%	41.8%	0.396
Prior year's income>$5000	74.5%	59.1%	41.8%	<0.001
Age	37.1	40.4	39.3	0.032
Proportion of prior quarters employed	0.665	0.772	0.373	<0.001
Average prior quarterly earnings	$3674	$4917	$1729	<0.001

Note: One observation per person. Proportion of quarters employed and average quarterly earnings are based on LWC records for 0.5 years to 2.5 years prior to the start of training. P-value for difference between groups based on ANOVA test.

time of the interview and nearly half having a prior year's income below $5000. The average post-training period quarterly earnings (averaging in zeroes for those who did not work during a quarter) was around $3500, implying an average annual earnings of around $14,000. Appendix Table A.1 reports the final two rows separated by employment group, and offers useful baselines to benchmark the change in outcomes by employment group later presented.

Table 4 examines the difference in observable baseline characteristics between the three employment groups. While there are no differences in treatment status probability or gender, there are large differences in prior labour outcomes and age. Specifically, those with a *job at start* have the highest probability of reporting a prior year's income above $5000, followed by the short-term unemployed and then the long-term unemployed. On the other hand, for the period of 0.5 to 2.5 years prior to training, the short-term unemployed had the highest proportion of quarters employed and the highest average quarterly earnings, followed by *job at start*. In all cases, *long-term unemployed* had the least favourable employment history. Thus, we find important differences between the short-term and long-term unemployed, and even potential evidence in favour of the potential for the short-term unemployed workers relative to the *job at start* group.

4. Methodology

In order to examine some of the potential mechanisms in the conceptual framework, we first explore the relationship between employment group and program outcomes (attendance and completion). In this case, we only use the treatment group and thus are not leveraging the randomization of the experiment, and are not necessarily estimating the causal impacts here. Eq. (1) shows the regression specification we use to examine this relationship. We regress outcome A_i, such as attendance in training round 1, for individual i on indicators for being in employment groups 2 or 3 (short or long-term unemployed), with the reference group of those who had jobs at start. We additionally control for baseline characteristics X_i (the fully interacted set of variables for gender, baseline annual income (less than or greater than $5000), and baseline age (younger or older than 35 years old)) and cohort fixed effects ψ_c. ε_i is the random error term

$$A_i = \alpha + \sum_{k=2,3} \beta_k Emp_{ik} + X_i\gamma + \psi_c + \varepsilon_i \tag{1}$$

For labour outcomes (employment and earnings), we leverage the RCT design and evaluate the impact of being randomly assigned to training while including several additional control variables to increase precision. Eq. (2) presents the model used.

$$Y_{it} = \alpha + \beta\, Treat_i + \lambda Y_{iB} + X_i\gamma + \psi_c + \phi_t + \varepsilon_{it} \tag{2}$$

Y_{it} is the dependant variable. It is either a binary variable indicating whether individual i is employed in quarter t, or a continuous variable giving the dollar amount of their earnings from all reported jobs in quarter t (including zeros for quarters the individual had no reported earnings). For each individual, we use all quarters after the end of their cohort's training period through the final quarter available, leading to multiple records across time for individuals. $Treat_i$ (whether or not individual i was randomly selected and invited for training) is the independent variable of interest (with independence ensured through the randomization). The covariate Y_{iB} represents the average outcome for the period 0.5 to 2.5 years prior to randomization, or in other words, at baseline. More specifically, it represents the fraction of quarters with at least one job as reported to LWC (for the regressions of employment status) or average quarterly earnings (for the regressions of earnings) from 2.5 years to 0.5 years prior to the randomization.

We additionally control for the baseline characteristics that defined the strata used for randomization (the fully interacted set of variables for gender, baseline employment status, baseline annual income (less than or greater than $5000), and baseline age (younger or older than 35 years old)), represented by X_i, and fixed effects for each cohort, represented by ψ_c. We also include time (year by quarter) fixed effects (ϕ_t). ε_{it} is the random error term.

We estimate Eq. (2) using OLS regression for the ITT model. As discussed in Section 2, individuals randomized into the control group were allowed to enter subsequent randomizations and potentially enter into treatment. We only include their treatment assignment from their first cohort to obtain identification of the effects free from selection bias. Thus, there are people in the control group that are later treated in a subsequent cohort. Removing them from the control group would create a non-random sample in the control group, as they are non-randomly deciding to re-enter later cohorts. They may be more motivated, which would yield an upwards bias on the treatment effect by removing more motivated control group members which would later on have better labour outcomes. On the other hand, they may have worse employment opportunities after being in the control group than those that opt not to return, which would downward bias the treatment effect by excluding them. The ultimate direction of the bias would be unclear; instead, we keep them in the sample with their original control group status. We do not have data on the extent to which study participants know each other outside of training. If those randomized to treatment are acquainted with those randomized to control, there is the potential for spill over of the learning from the treatment to control, which would dampen the estimated effects.

However, as noted in Table 3, there is non-compliance in both directions. The allowance of individuals to enter later treatment groups and our modelling decision to classify individuals based on their first assignment will reduce the ITT estimates from what it would have been if such re-entry was disallowed. To address this, we also estimate a LATE model for attendance in at least one session of training using two-stage least squares regressions, with treatment assignment as the instrument. This allows us to account for the fact that there is non-compliance in both directions (treatment persons never attending, and control persons entering later treatment groups). Attendance is endogenous to the trainees' labour market opportunities and unobserved motivation and ability. However, initial treatment assignment, given that it is random, will satisfy the independence assumption. For example, programme participant's earnings should not be a function of the random assignment to treatment except through their propensity to attend the training and derive the hypothesized benefits therefrom. If an individual randomly assigned to the treatment group does not attend even once, we expect them to have the same wage outcome as a control group person who in the counterfactual of a training offer would likewise have not attended. For the strong instrument assumption, the first stage F-statistics are 39.7 and 39. for employment and earnings, respectively.

We generate non-response weights to account for the 28 subjects with SSNs that we were not able to match to the LWC records. We fit a logistic regression of non-missing (matched SSN) status onto the person's treatment status and the four randomization strata of baseline age, sex, income, and employment. Using this fitted regression, we predict the probability of being a non-missing response and generate the regression weights as the inverse of this predicted probability.

We also explore treatment effect heterogeneity along the dimension of their employment group at randomization. To do so, we estimate Eq. (3). Emp_{ik} is an indicator variable for individual i being in employment group k=1 (*job at start*), 2 (*short-term unemployed*), or 3 (*long-term unemployed*). This allows us to explore how the treatment effects (ITT and LATE) differ depending on the baseline employment group.

$$Y_{it} = \alpha + \sum_{k=1,2,3} \beta_k Treat_i \times Emp_{ik} + \lambda Y_{iB} + X_i\gamma + \psi_c + \phi_t + \varepsilon_{it} \tag{3}$$

Finally, we examine what industries individuals end up working in, with the goal of examining if trainees are more likely to work in the target industries. To do so, we use Eq. (2) again, where the outcome Y_{it} is whether they work in a given 3-digit NAICS code; we separately estimate the regression for each of the 3-digit NAICS codes in which at least one worker is employed in our analytic sample. We control for the same covariates as for employment and earnings outcomes, including an indicator for them having worked in that given industry in the observed past of 0.5 to 2.5 years before training (Y_{iB}). We also again include all individuals in this regression, including those who are not working (and thus take on a value of zero for the dependant variable). We do this across all NAICS codes present in the data, sort the coefficients by size, and order and report the industries and coefficients as a measure of the impact of training on target employing industry. We do this separately by training pathway, given they have separate target industries. We also calculate and present t-statistics

For Eq. (1), we use Eicker-Huber White heteroskedasticity-robust standard errors. For analyses of Eqs. (2) and (3), we cluster the standard errors by cohort crossed with treatment status, to reflect expected intraclass correlations.

5. Results

5.1. Differences in attendance and completion by employment group

Table 5 examines how training attendance and completion rates differ by employment group. We find that there are not large differences between groups, with the exception of the long-term unemployed being significantly more likely to attend and complete the second round of training. This is consistent with the idea that those in the long-term unemployed group have lower opportunity costs of their time, perhaps due to on-average worse employment offers. This would potentially decrease treatment effects for this group.

5.2. Effects on employment and earnings

We present the overall findings of the programme on the likelihood of being employed in Table 6. Pooling employment groups, we find no significant effect on employment, although the results are positive and economically relevant. The LATE estimate for those who attended at least one class of the first round of training is larger than the ITT estimate, but also is not statistically significant.

When we examine treatment effect heterogeneity by baseline employment status, the only group with statistically significant treatment effects for employment probability are the short-term unemployed workers. The 14 percentage point higher likelihood of employment is not only statistically significant, but large in magnitude, especially compared to the control group short-term unemployed average of 60.8 % (Appendix Table A.1). This means the programme is helping one group of participants that policy makers may be seeking to help.

Table 7 presents the treatment effects for quarterly earnings. We find marginally significant pooled ITT and LATE estimates. Given the control group average of $3567 (Table 3), this is a 12 % increase for the ITT and a 20 % increase for the LATE. When examining the effect by baseline employment group, we again find that the largest treatment effect is for the short-term unemployed workers. The treatment effects are large too, with $1472 ITT representing over a 50 % increase over the control group short-term unemployed average ($2830, Appendix Table A.1), and a LATE that is nearly double the same control group average. However, now the long-term unemployed group also had large and significant ITT and LATE estimates; compared to the control group long-term unemployed average quarterly earnings of $2333, it is proportionally even larger, with the LATE representing over a doubling of the control group average.

Table 8 explores the extent to which treatment effects on employment and earnings vary by training pathway. Trainees in the health

Table 5
Regression results for training attendance and completion.

	Attended training 1	Completed training 1	Attended training 2	Completed training 2	Obtained credential
Short-term	-0.004	-0.051	0.016	0.047	-0.052
unemployed	(0.068)	(0.076)	(0.045)	(0.045)	(0.074)
Long-term	-0.058	0.109	0.108*	0.144**	0.031
unemployed	(0.076)	(0.084)	(0.057)	(0.056)	(0.092)
Observations	197	197	197	197	197
R-squared	0.231	0.358	0.629	0.547	0.405
Sample mean	0.822	0.640	0.223	0.178	0.513

Note: Reference group contains those who have a job at randomization. Regressions additionally control for fixed effects for cohort, and randomization strata (gender X age over 35 X baseline employment X baseline income over $5000).

Table 6
Regression results for being employed.

	ITT		LATE	
	(1)	(2)	(3)	(4)
Treat	0.035		0.057	
	(0.026)		(0.039)	
Treat X job at start		-0.030		-0.040
		(0.038)		(0.056)
Treat X short-term		0.140**		0.221**
unemployed		(0.064)		(0.091)
Treat X long-term		0.038		0.080
unemployed		(0.065)		(0.093)
Avg. pre-randomization	0.252***	0.251***	0.252***	0.239***
employment	(0.068)	(0.076)	(0.065)	(0.076)
Observations	1901	1901	1901	1901
R-squared	0.114	0.121	0.118	0.116
Mean for control group	0.639	0.639	0.621	0.621
p-value for Wald test of equality of treatment effects between employment groups				
Group 1 versus 2		0.035		0.022
Group 1 versus 3		0.403		0.308
Group 2 versus 3		0.304		0.239

Note: ITT is the intent to treat effect. LATE is the local average treatment effect of attendance at training for compliers, using random treatment assignment as an instrumental variable. Regressions additionally control for fixed effects for cohort, year-by-quarter, and randomization strata (gender X age over 35 X baseline employment X baseline income over $5000). Employment group 1: job at start; employment group 2: short-term unemployed; employment group 3: long-term unemployed. *p<.10, **p<.05, ***p<.01 with standard errors clustered by cohort/treatment assignment.

care have large and significant effects for both outcomes, while advanced manufacturing trainees only have statistically significant effects for quarterly earnings. Program participants in information technology pathway cohorts have no statistically significant treatment effects.

5.3. Effects on industry of employment

We next turn our attention to the industries that treated individuals are most (and least) likely to work in compared to the control group. We group industry at the three-digit level of NAICS codes. Of the 99 industries, individuals in the advanced manufacturing pathway (treatment or control) worked in 47 of them. Individuals in health care pathway worked in 29 of the 99 industries, and those in the IT pathway worked in 36 of the 99 industries. While we estimated the difference between treatment and control for all industries, we only report the top and bottom three industries in terms of size of the coefficient, reported in Table 8. Appendix Tables A2–A4 report it for all industries. In Table 9, we show some evidence that the training is leading trainees to be more likely to be employed in the intended target industry, which we interpret as evidence that the training is developing industry-specific human capital or building connections helpful for finding these jobs (Neal, 1995). Recall that both the treatment and control group are comprised of individual who self-selected into the training area, and presumably are interested in working in the target industries.

For advanced manufacturing, the top difference industry is speciality trade contractors, where the treatment group is five percentage points more likely to work. This industry is related to the goals of the training. For health care, the effect of treatment is larger, as the treatment group is over 30 percentage points more likely to work in the hospital industry than the control group. Also encouraging for the health care pathway are the negative treatment effects: in the health care pathway, the training assignment is causing them to be less likely to work in administrative and support services as well as the food services and dining places industry. Thus, they are moving out of jobs that are more likely to be low-skilled and lower paying and into a better career path. We see no such connections for the IT pathway, aligning with the null results on employment and earnings. Many industries employ people in information technology occupations so these findings should be interpreted with caution.

The results in Table 8 are influenced by the size of the industries. That is, an industry could have a larger coefficient because the industry itself is larger. In Appendix Tables A2–A4, we present the full list of industries observed, and provide two alternatives for rankings, for each of the pathways. The first column contains the coefficients used for ranking in Table 8. The next column divides the difference coefficients in Table 8 by the percent of the Orleans Parish workforce in that industry (from the 2017 QCEW), providing the percent increase in the probability relative to the unconditional probability of working in an in-

Table 7
Regression results for quarterly earnings.

	ITT		LATE	
	(1)	(2)	(3)	(4)
Treat	435.990*		705.144*	
	(256.708)		(389.069)	
Treat X job at start		-567.055		-804.269
		(473.477)		(676.460)
Treat X short-term unemployed		1471.771**		2514.762***
		(568.496)		(921.909)
Treat X long-term unemployed		1414.467***		2499.909***
		(405.167)		(776.785)
Avg. pre-randomization earnings	0.338***	0.349***	0.336***	0.347***
	(0.089)	(0.091)	(0.088)	(0.091)
Observations	1901	1901	1901	1901
R-squared	0.204	0.218	0.205	0.191
Mean for control group	3567	3567	3411	3411
p-value for Wald test of equality of treatment effects between employment groups				
Group 1 versus 2		0.023		0.014
Group 1 versus 3		0.004		0.005
Group 2 versus 3		0.935		0.989

Note: ITT is the intent to treat effect. LATE is the local average treatment effect of attendance at training for compliers, using random treatment assignment as an instrumental variable. Regressions additionally control for fixed effects for cohort, year-by-quarter, and randomization strata (gender X age over 35 X baseline employment X baseline income over \$5000). Quarterly earnings are US dollars normalized to 2018. Employment group 1: job at start; employment group 2: short-term unemployed; employment group 3: long-term unemployed. *p<.10, **p<.05, ***p<.01 with standard errors clustered by cohort/treatment assignment.

Table 8
Regression results by pathway.

	Employment		Quarterly Earnings	
	ITT	LATE	ITT	LATE
Advanced Manufacturing	-0.003	-0.004	451.173**	759.030**
	(0.027)	(0.043)	(169.518)	(300.106)
Observations	955	955	955	955
R-squared	0.089	0.088	0.225	0.233
Mean for control group	0.620	0.584	2788	2776
Health care	0.151***	0.185***	1739.929***	2150.268***
	(0.013)	(0.026)	(148.055)	(174.013)
Observations	371	371	371	371
R-squared	0.299	0.282	0.400	0.361
Mean for control group	0.681	0.694	3409	3766
Information Technology	0.019	0.034	-708.513	-1211.532
	(0.049)	(0.085)	(785.601)	(1273.747)
Observations	575	575	575	575
R-squared	0.176	0.172	0.238	0.228
Mean for control group	0.642	0.652	4773	4369

Note: ITT is the intent to treat effect. LATE is the local average treatment effect of attendance at training for compliers, using random treatment assignment as an instrumental variable. Regressions additionally control for fixed effects for cohort, year-by-quarter, and randomization strata (gender X age over 35 X baseline employment X baseline income over \$5000). Quarterly earnings are US dollars normalized to 2018. *p<.10, **p<.05, ***p<.01 with standard errors clustered by cohort/treatment assignment.

Table 9
Largest and smallest differences between treatment and control industry of employment after end of training period (NAICS 3-Digit Codes).

	Cohort's target industry		
	Advanced manufacturing	Health care	Information technology
top 3	Speciality trade contractors (4.9***)	Hospitals (32.2***)	Food Services and Drinking Places (5.8***)
	Professional, Scientific, and Technical Services (2.8***)	Social Assistance (2.9)	Justice, Public Order, and Safety Activities (4.3**)
	Food Services and Drinking Places (2.1)	Educational Services (2.4**)	Couriers and Messengers (3.8***)
Bottom 3	Educational Services (-2.7***)	Administrative and Support Services (-5.9**)	Performing Arts, Spectator Sports, and Related Industries (-5.0***)
	Heavy and Civil Engineering Construction (-2.7***)	Food Services and Drinking Places (-5.8***)	Motion Picture and Sound Recording Industries (-4.2***)
	Personal and Laundry Services (-1.7***)	Furniture and Home Furnishings Stores (-4.3***)	Health and Personal Care Stores (-3.9***)

Note: ITT model. Each cell is based on a separate regression. The regressions additionally control for having worked in the industry before randomization, gender, age, baseline employment status, and baseline income, as well as cohort and time fixed effects. Treatment effects are in parentheses for the probability of working in that industry (on a scale from 0 to 100). *p<.10, **p<.05, ***p<.01 with standard errors clustered by cohort/treatment assignment.

dustry. The final column presents the t-statistic for the coefficient from the regression, which instead scales the difference by the standard error. While there are some differences in the rankings between the three columns, they generally are consistent, with for example speciality trade contractors being ranked first in all three measures for advanced manufacturing, and hospitals being listed first all three measures for health care.

Appendix Table A.5 replicates Table 8 but for each of the three employment groups, done in separate regressions. The results suggest that the training programs provided specific human capital regardless of baseline employment status for the healthcare pathway - hospitals lead the list for all three employment categories. At the other extreme, none of the baseline employment categories show movements to related industries for the IT pathway, suggesting that the program was uniformly unsuccessful in placing individuals into the target industry. The advanced manufacturing pathway moved people into related industries for both those who had a job and those who were short-term unemployed, but not for the long term unemployed. This suggests trainees in the advanced manufacturing pathway required recent labour market experience to make use of their training.

6. Discussion

There is a persistent need to consider ALMPs that would help disadvantaged persons struggling with employment. Amongst the many options are programmes that work to develop the human capital of the individuals for skills that are in demand in the labour market. Job training programmes have been shown to have promise, but there is still a need to understand how effective they are and what can help the segments of the population that are more vulnerable, including those without any job. In this paper, we evaluate a job training programme in New Orleans—Career Pathways—aimed at disadvantaged populations unsatisfied with their employment situation, and investigate for which sub-populations of baseline employment status the programme works. We evaluate an RCT to understand how effective this 2-to-4 month training programme was across 20 different training cohorts in advanced manufacturing, health care, and information technology. We find a meaningful intent to treat impact on earnings which represented over a 20 % increase, or over $400 per quarter of a year. This impact nearly doubled when we accounted for training attendance using a LATE estimator.

We find that people who are working at randomization had the least favourable impact of training. We speculate that there could be several reasons for this low impact. People who are working could either quit their current job to focus on training or try to continue working during training. Either choice could present challenges. For those who quit, they are faced with the challenge of replacing that wage income or subsisting on less income, which places pressures on the trainee and their dependents. These pressures may lead to those trainees quitting the training programme to replace their lost income immediately, albeit without the skills that come with completed training. Those who keep working may find it difficult to attend to the rigors of training, causing them to either drop out or give it less than their complete attention. Additionally, those with jobs at randomization are likely to have opportunities that arise in the form of advancement on their current job or new job offers (Schönberg, 2007; Pissarides and Wadsworth, 1994; Boswell et al., 2012). These opportunities will accrue to both those in the training and in the control groups. This means the contrast with members of the control group will be lower. It also means that increasing opportunities on the current job for trainees will add to the pressure to stop training and focus on the current job.

The largest treatment effects were for those who were short-term unemployed entering training, and then for those long-term unemployed. Our finding that the training programme had a larger impact on people who had a job recently (short-term unemployed) than on people who had not worked in more than six months suggests that there is an interaction between having worked recently and training. This could reflect factors on either the trainee side or the potential employer side. Those who have worked recently might be more able to adjust to the scheduling and effort demands of job training. Employers might prefer applicants with new skill acquisition only when they have a recent work history.

While the overall magnitudes of these effects are roughly in line with prior experimental and quasi-experimental estimates of other recent job training programs, our finding that the individuals who benefited most were short-term unemployed has important implications. This aligns with the Biewen et al. (2014)'s findings in a quasi-experimental evaluation in Germany and forms an interesting and important target population for these kinds of job training programs. Local governments could consider this type of programme (with the screening and demand-driven features) for their Dislocated Workers programmes as well, to see if such a model would improve the success of the training programmes which have typically been found to be less successful (Andersson et al., 2013). Although we cannot definitively state why this programme was so successful especially for short-term unemployed individuals who entered the programme, one reason could be because of the key element of screening individuals before admitting them to training. Specifically, OWD verified that the individuals had sufficient literacy and numeracy through the Test of Adult Basic Education as well as other screening exercises. Such a model could be tested in the Dislocated Worker program to see if that improves the effectiveness, where historically the program has struggled more to produce positive results.

Another reason for the success here may have been the demand-driven nature of the programmes. Prior evaluations of Dislocated Worker training programmes have been for WIA programmes that were created prior to the WIOA requirement to "emphasize sector-based strategies to promote employment in high-demand industries and occupations" (Fortson et al., 2017, p. 5).

We find strong evidence that individuals assigned to training in two of the three pathways (advanced manufacturing and health care) were more likely to end up working in the related target industries than their comparison workers in the control group, comporting with those pathways having statistically significant treatment effects on quarterly earnings (and on employment, for health care). There is further evidence that this was at least partially driven by movement out of low-skill industries, suggesting that the development of human capital through the training programme allowed them to move into better careers in the target industries (Neal, 1995). This stands in contrast to Fortson et al. (2017) who found no increased probability of working in the occupations for which they were trained.

Overall, we find the results of the analysis highly encouraging. Career Pathways was able to succeed with the population of high interest, those without jobs at start of training, especially the short-term unemployed workers. And that success was accounted for along several dimensions, including higher employment rates, earnings, and eventual employment in higher skill industries from which they can make careers.

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Appendix

Table A1
Unadjusted outcomes by employment group.

	Control	Treatment	Difference	Std. Error of Diff.	P-value	Effect Size
Employed						
Job at start	0.710	0.717	-0.006	0.059	0.913	0.014
Short-term unemployed	0.608	0.734	-0.126	0.066	0.065	0.269
Long-term unemployed	0.510	0.556	-0.047	0.089	0.603	0.093
Quarterly earnings						
Job at start	4466	4044	423	691	0.544	0.099
Short-term unemployed	2830	4075	-1245	593	0.043	0.318
Long-term unemployed	2333	3311	-978	608	0.118	0.253

Table A2
Advanced manufacturing differences between treatment and control industry of employment after end of training period (NAICS 3-Digit Codes).

	Diff/pop.		
Industry	Diff.	rate	t-stat
Speciality trade contractors (1, 1, 1)	4.94	4.81	4.84
Professional, Scientific, and Technical Services (2, 14, 2)	2.83	0.35	4.08
Food Services and Drinking Places (3, 18, 12)	2.12	0.14	1.32
General Merchandise Stores (4, 7, 4)	1.92	1.28	2.52
Accommodation (5, 15, 11)	1.43	0.23	1.33
Justice, Public Order, and Safety Activities (6, 11, 3)	1.43	0.48	2.98
Administrative and Support Services (7, 17, 20)	1.06	0.17	0.44
Construction of Buildings (8, 5, 13)	0.91	1.61	1.29
Management of Companies and Enterprises (9, 12, 7)	0.76	0.45	1.52
Truck Transportation (10, 4, 10)	0.59	1.76	1.33
Water Transportation (11, 3, 5)	0.52	1.84	2.04
Utilities (12, 2, 6)	0.51	4.65	1.76
Ambulatory Health Care Services (13, 20, 17)	0.39	0.11	0.95
Scenic and Sightseeing Transportation (14, 6, 8)	0.35	1.40	1.42
Waste Management and Remediation Services (15, 8, 19)	0.34	1.21	0.74
Sporting Goods, Hobby, Musical Instrument, and Book Stores (16, 9, 14)	0.22	1.20	1.00
Wholesale Electronic Markets and Agents and Brokers (17, 10, 15)	0.21	0.73	1.00
Couriers and Messengers (18, 13, 9)	0.20	0.43	1.39
Transit and Ground Passenger Transportation (19, 16, 18)	0.17	0.20	0.86
Real Estate (20, 19, 21)	0.15	0.14	0.22
Amusement, Gambling, and Recreation Industries (21, 21, 16)	0.09	0.10	0.98
Motion Picture and Sound Recording Industries (22, 22, 22)	-0.02	-0.01	-0.02
Merchant Wholesalers, Nondurable Goods (23, 24, 23)	-0.03	-0.04	-0.07
Food Manufacturing (24, 26, 25)	-0.04	-0.05	-0.09
Support Activities for Transportation (25, 25, 24)	-0.05	-0.04	-0.08
Hospitals (26, 23, 26)	-0.11	-0.02	-0.18
Fabricated Metal Product Manufacturing (27, 41, 30)	-0.17	-2.12	-1.00
Gasoline Stations (28, 32, 28)	-0.21	-0.67	-0.99
Performing Arts, Spectator Sports, and Related Industries (29, 27, 29)	-0.21	-0.12	-1.00
Clothing and Clothing Accessories Stores (30, 29, 31)	-0.28	-0.22	-1.26
Repair and Maintenance (31, 34, 36)	-0.30	-0.74	-1.63
Motor Vehicle and Parts Dealers (32, 35, 33)	-0.38	-0.80	-1.42
Social Assistance (33, 28, 32)	-0.43	-0.18	-1.29
Miscellaneous Manufacturing (34, 43, 37)	-0.56	-3.71	-1.74
Nursing and Residential Care Facilities (35, 31, 27)	-0.61	-0.39	-0.60
Health and Personal Care Stores (36, 36, 40)	-0.73	-0.95	-2.02
Miscellaneous Store Retailers (37, 37, 34)	-0.74	-0.97	-1.52
Museums, Historical Sites, and Similar Institutions (38, 38, 38)	-0.77	-1.01	-1.77
Merchant Wholesalers, Durable Goods (39, 39, 35)	-0.95	-1.18	-1.63
Printing and Related Support Activities (40, 46, 42)	-1.13	-9.12	-2.31
Food and Beverage Stores (41, 33, 41)	-1.20	-0.71	-2.05
Furniture and Home Furnishings Stores (42, 45, 43)	-1.33	-4.28	-2.53
Building Material and Garden Equipment and Supplies Dealers (43, 42, 39)	-1.41	-2.58	-1.82
Textile Product Mills (44, 47, 45)	-1.52	-49.42	-2.89
Personal and Laundry Services (45, 40, 44)	-1.74	-1.20	-2.72
Heavy and Civil Engineering Construction (46, 44, 46)	-2.65	-4.20	-3.20
Educational Services (47, 30, 47)	-2.75	-0.24	-3.72

Table A3
Health care differences between treatment and control industry of employment after end of training period (NAICS 3-Digit Codes).

Industry	Diff.	Diff/pop. rate	t-stat
Hospitals (1, 1, 1)	32.20	5.84	9.10
Social Assistance (2, 3, 4)	2.89	1.23	1.48
Educational Services (3, 8, 2)	2.42	0.21	2.11
Professional, Scientific, and Technical Services (4, 10, 8)	1.21	0.15	0.64
Insurance Carriers and Related Activities (5, 2, 3)	1.16	1.41	1.74
Support Activities for Transportation (6, 4, 5)	1.02	0.91	1.41
Ambulatory Health Care Services (7, 7, 9)	1.00	0.28	0.43
Food and Beverage Stores (8, 6, 6)	0.48	0.28	0.99
Health and Personal Care Stores (9, 5, 7)	0.35	0.46	0.99
Credit Intermediation and Related Activities (10, 9, 10)	0.27	0.16	0.21
Management of Companies and Enterprises (11, 11, 11)	0.08	0.05	0.08
Miscellaneous Store Retailers (12, 14, 13)	-0.36	-0.48	-0.98
Gasoline Stations (13, 21, 14)	-0.42	-1.37	-0.98
Wholesale Electronic Markets and Agents and Brokers (14, 23, 16)	-0.43	-1.49	-1.00
Couriers and Messengers (15, 18, 15)	-0.49	-1.07	-0.99
Rental and Leasing Services (16, 22, 17)	-0.50	-1.44	-1.00
Private Households (17, 28, 21)	-0.72	-5.10	-1.42
Electronics and Appliance Stores (18, 27, 22)	-0.90	-4.62	-1.43
Building Material and Garden Equipment and Supplies Dealers (19, 26, 20)	-1.01	-1.85	-1.41
Accommodation (20, 12, 12)	-1.03	-0.17	-0.58
Amusement, Gambling, and Recreation Industries (21, 19, 23)	-1.27	-1.32	-1.72
Nursing and Residential Care Facilities (22, 16, 18)	-1.41	-0.91	-1.11
Justice, Public Order, and Safety Activities (23, 15, 19)	-1.63	-0.55	-1.12
Clothing and Clothing Accessories Stores (24, 20, 25)	-1.68	-1.33	-2.00
Performing Arts, Spectator Sports, and Related Industries (25, 24, 24)	-2.56	-1.50	-1.91
General Merchandise Stores (26, 25, 26)	-2.70	-1.79	-2.08
Furniture and Home Furnishings Stores (27, 29, 29)	-4.26	-13.69	-3.41
Food Services and Drinking Places (28, 13, 28)	-5.81	-0.40	-2.83
Administrative and Support Services (29, 17, 27)	-5.85	-0.93	-2.12

Table A4
Information technology differences between treatment and control industry of employment after end of training period (NAICS 3-Digit Codes).

Industry	Diff.	Diff/pop. rate	t-stat
Food Services and Drinking Places (1, 15, 3)	5.78	0.39	3.04
Justice, Public Order, and Safety Activities (2, 11, 11)	4.28	1.43	2.01
Couriers and Messengers (3, 1, 1)	3.77	8.18	3.69
Nursing and Residential Care Facilities (4, 6, 2)	3.70	2.39	3.20
Food Manufacturing (5, 4, 6)	3.07	3.68	2.68
Amusement, Gambling, and Recreation Industries (7, 7, 5)	2.18	2.27	2.69
Real Estate (6, 8, 4)	2.18	1.96	2.69
Building Material and Garden Equipment and Supplies Dealers (8, 5, 7)	1.90	3.46	2.46
General Merchandise Stores (9, 13, 13)	1.89	1.26	1.48
wholesale electronic markets and agents and brokers (10, 2, 8)	1.57	5.43	2.25
Speciality trade contractors (11, 12, 12)	1.42	1.39	1.49
Truck Transportation (12, 3, 10)	1.38	4.15	2.04
Clothing and Clothing Accessories Stores (13, 14, 9)	1.30	1.03	2.19
Motor Vehicle and Parts Dealers (14, 9, 14)	0.76	1.62	1.39
Ambulatory Health Care Services (15, 16, 16)	0.29	0.08	0.99
Sporting Goods, Hobby, Musical Instrument, and Book Stores (16, 10, 15)	0.28	1.55	1.01
Merchant Wholesalers, Nondurable Goods (17, 17, 17)	0.01	0.01	0.17
Rental and Leasing Services (18, 22, 18)	-0.07	-0.21	-0.06
Accommodation (19, 18, 20)	-0.41	-0.07	-0.35
Nonstore Retailers (20, 35, 25)	-0.44	-6.89	-1.38
Construction of Buildings (21, 25, 26)	-0.45	-0.79	-1.39
Personal and Laundry Services (22, 24, 24)	-0.57	-0.39	-1.37
Hospitals (23, 19, 22)	-0.57	-0.10	-0.78
Administrative and Support Services (24, 20, 19)	-0.67	-0.11	-0.31
Merchant Wholesalers, Durable Goods (25, 27, 21)	-0.93	-1.15	-0.75
Repair and Maintenance (26, 31, 30)	-1.05	-2.59	-1.92
Insurance Carriers and Related Activities (27, 28, 27)	-1.09	-1.32	-1.72
Telecommunications (28, 36, 32)	-2.06	-17.02	-2.70
Educational Services (29, 21, 23)	-2.20	-0.19	-1.12
Professional, Scientific, and Technical Services (30, 23, 29)	-2.38	-0.30	-1.90
Social Assistance (31, 26, 28)	-2.44	-1.04	-1.76
Credit Intermediation and Related Activities (32, 29, 31)	-2.52	-1.53	-2.68
Support Activities for Transportation (33, 30, 33)	-2.66	-2.37	-3.00
Health and Personal Care Stores (34, 34, 36)	-3.88	-5.08	-3.71
Motion Picture and Sound Recording Industries (35, 33, 34)	-4.16	-3.40	-3.44
Performing Arts, Spectator Sports, and Related Industries (36, 32, 35)	-4.98	-2.92	-3.45

Table A5
Largest and smallest differences between treatment and control industry of employment after end of training period (NAICS 3-Digit Codes) by employment group.

		Cohort's target industry		
		Advanced manufacturing	Health care	Information technology
Had job at start	top 3	Speciality trade contractors (6.6***)	Hospitals (24.0***)	Food Services and Drinking Places (9.4***)
		Administrative and Support Services (2.6)	Ambulatory Health Care Services (6.3)	Nursing and Residential Care Facilities (5.5***)
		Management of Companies and Enterprises (2.6***)	Educational Services (5.3**)	Administrative and Support Services (5.2**)
	Bottom 3	Educational Services (-5.2***)	Food Services and Drinking Places (-10.3***)	Performing Arts, Spectator Sports, and Related Industries (-7.3***)
		Textile Product Mills (-3.6***)	Furniture and Home Furnishings Stores (-6.4***)	Health and Personal Care Stores (-6.0***)
		Merchant Wholesalers, Durable Goods (-3.3***)	Performing Arts, Spectator Sports, and Related Industries (-5.8*)	Credit Intermediation and Related Activities (-3.8***)
Short-term unemployed	Top 3	Professional, Scientific, and Technical Services (5.8***)	Hospitals (46.5***)	Amusement, Gambling, and Recreation Industries (12.8***)
		Justice, Public Order, and Safety Activities (5.0***)	Professional, Scientific, and Technical Services (7.6)	Real Estate (12.8***)
		Support Activities for Transportation (5.0**)	Nursing and Residential Care Facilities (2.2)	Justice, Public Order, and Safety Activities (10.2***)
	Bottom 3	Administrative and Support Services (-10.9**)	Administrative and Support Services (-12.2***)	Support Activities for Transportation (-16.4***)
		Personal and Laundry Services (-3.9*)	General Merchandise Stores (-7.3**)	Motion Picture and Sound Recording Industries (-12.0***)
		Health and Personal Care Stores (-3.7**)	Credit Intermediation and Related Activities (-5.6**)	Social Assistance (-11.1***)
Long-term unemployed	Top 3	Food Services and Drinking Places (7.9**)	Hospitals (41.2***)	Motion Picture and Sound Recording Industries (1.1)
		Speciality trade contractors (5.7**)	Social Assistance (6.5*)	Administrative and Support Services (0.4)
		Professional, Scientific, and Technical Services (5.0**)	Administrative and Support Services (3.8)	Performing Arts, Spectator Sports, and Related Industries (0.1)
	Bottom 3	Furniture and Home Furnishings Stores (-8.8***)	Justice, Public Order, and Safety Activities (-12.5**)	Professional, Scientific, and Technical Services (-39.2**)
		Administrative and Support Services (-7.4)	Management of Companies and Enterprises (-11.9**)	General Merchandise Stores (-32.3*)
		Building Material and Garden Equipment and Supplies Dealers (-2.7**)	Professional, Scientific, and Technical Services (-5.6*)	Social Assistance (-10.6)

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